$$\frac{du_1}{dt} = -u_1 + 2u_2 \qquad u(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\frac{du_2}{dt} = u_1 - 2u_2$$

$$A = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}$$

$\lambda = 0, -3$

$\lambda_1 = 0 \qquad \lambda = -3$

$x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$Ax_1 = 0x_1 \qquad Ax_2 = -3x_2$

Solution:

$$u(t) = c_1 e^{\lambda_1 t} x_1 + c_2 e^{\lambda_2 t}$$

$$= c_1 \cdot 1 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

At $t = 0$;

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$u(t) = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{1}{3} e^{-3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

At $t = \infty$, $\rightarrow$ steady state

$$u(\infty) = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(1) Stability

$u(t) \rightarrow 0$

$e^{\lambda t} \rightarrow 0 \quad \text{Re}\,\lambda < 0$

(2) Steady State

$\lambda_1 = 0$ & other $\text{Re}\,\lambda < 0$

(3) Blow up

Any $\text{Re}\,\lambda > 0$

$$\frac{du}{dt} = Av$$

$$\frac{S\,dv}{dt} = ASv$$

$$\frac{dv}{dt} = S^{-1}ASv$$
$$= \Lambda v$$

S is eigen-matrix (P)

Λ is eigen-value diagonal matrix (D)

$$v(t) = e^{\Lambda t} v(0)$$

$$u(t) = Se^{\Lambda t}S^{-1}u(0)$$
$$= e^{At}u(0)$$

$$e^{At} = Se^{\Lambda t}S^{-1}$$

Matrix Exponentials:

$$e^{At} = I + At + \frac{(At)^2}{2} + \frac{(At)^3}{6} + \dots + \frac{(At)^n}{n!}$$

$$(I - At)^{-1} = I + At + (At)^2 + \dots + (At)^n$$

$$e^{At} = I + S\Lambda S^{-1}t + \frac{S\Lambda^2 S^{-1}t^2}{2} + \dots$$

$$= SS^{-1} + S\Lambda S^{-1}t + \frac{S\Lambda^2 S^{-1}t^2}{2} + \dots$$

$$= S\left(I + \Lambda t + \frac{\Lambda^2}{2}t^2 + \dots\right)S^{-1}$$

$$= Se^{\Lambda t}S^{-1}$$ } A must dialigonizable for this to work